

FINITELY MANY MASS POINTS ON THE LINE UNDER THE INFLUENCE
OF AN EXPONENTIAL POTENTIAL -- AN INTEGRABLE SYSTEM

Jürgen Moser*

Courant Institute of Mathematical Sciences, NYU, New York 10012

1. Analogue of the Toda Lattice for Finitely Many Mass Points

We consider the analogue of the Toda lattice [8] where only a finite number of mass points are admitted which move freely on the real axis. Denoting the position of the mass points by x_k , $k = 1, \dots, n$, we form the Hamiltonian

$$(1.1) \quad H = \frac{1}{2} \sum_{k=1}^n y_k^2 + \sum_{k=1}^{n-1} e^{(x_k - x_{k+1})}$$

with the differential equations

$$(1.2) \quad \begin{aligned} \dot{x}_k &= H_{y_k} = y_k, & k &= 1, 2, \dots, n \\ \dot{y}_k &= -H_{x_k} = e^{x_{k-1} - x_k} - e^{x_k - x_{k+1}}, & k &= 2, \dots, n-1 \\ \dot{y}_1 &= -H_{x_1} = -e^{x_1 - x_2} \\ \dot{y}_n &= -H_{x_n} = e^{x_{n-1} - x_n} \end{aligned}$$

Thus we can write our system (1.2) as

$$(1.2') \quad \dot{x}_k = e^{x_{k-1} - x_k} - e^{x_k - x_{k+1}}, \quad k = 1, \dots, n$$

if we set $e^{x_0 - x_1} = 0$ and $e^{x_n - x_{n+1}} = 0$, that is we have the formal boundary condition

$$(1.3) \quad x_0 = -\infty, \quad x_{n+1} = +\infty.$$

It is the aim to study completely the flow determined by this

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system of differential equations and relate the solution to the existence of n integrals of the motion. These integrals are essentially the same as those found by Henon [4] and Flaschka [1] for the same system of differential equations (1.2') under periodic boundary conditions, say

$$(1.3') \quad \begin{aligned} x_{k+n} &= x_k + 1 \\ y_{k+n} &= y_k \end{aligned} \quad k = 0, \pm 1, \dots$$

The crucial difference between the two problems is that the boundary condition (1.3') gives rise to a compact energy surface and the solutions are expected to be quasiperiodic, lying on tori, as one is familiar from integrable Hamiltonian systems. If we impose the boundary condition (1.3) instead of (1.3') the energy surface is noncompact, as the particles can run to infinity. In fact, we will show, as is intuitively clear, that for any initial configuration mutual distances between all particles grow indefinitely, i.e.

$x_{k-1} - x_k \rightarrow \infty$ for $k = 2, \dots, n$; and they behave asymptotically like free particles depending linearly on time. This suggests the scattering problem: To determine the relation between this asymptotic motion for the past and the future. This can be done explicitly here and one finds that $y_{n-k+1}(+\infty) = y_k(-\infty)$, so that at $t = +\infty$ the first particle has the velocity of the last at $t = -\infty$ etc. as in a familiar experiment of collision of steel balls. Moreover, the phase relation can also be determined explicitly and we will show that

$$x_{n-k+1}(t) - x_k(-t) - 2y_k^- t + \sum_{j < k} \log (y_j^- - y_k^-)^2 = \sum_{j > k} \log (y_j^- - y_k^-)^2,$$

where $y_j^- = y_j(-\infty)$ are assumed ordered according to size. Thus the particles behave asymptotically as if they interacted just pairwise! This will be derived in Section 4.